

An exact cycle atlas for multiway perfect shuffles

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Abstract

We treat a perfect multiway shuffle as a deterministic finite dynamical system. For integers $k \geq 2$ and $n \geq 1$, the kn card positions are the nonzero residues modulo $M = kn + 1$ and one tick is $\rho_{k,n}(i) = ki \pmod{M}$. A fixed-point congruence gives $\text{Fix}(\rho^r) = \gcd(k^r - 1, M) - 1$; gcd strata give each position's least period, and Möbius inversion separates primitive cycles from repeats. We prove the resulting formulas for the whole parameter family, verify them against literal packet interleaving on 50 parameter pairs, and factor the source-local finite zeta and Koopman characteristic polynomial. The atlas is exact but deliberately makes no target arithmetic or Hilbert–Pólya claim.

1 Frozen shuffle and packet convention

Fix $k \geq 2$, $n \geq 1$, put $M = kn + 1$, and let $D_M = \{1, \dots, M-1\}$. We use the in-shuffle convention of Ellis–Fan–Shallit: k equal piles are interleaved in reverse pile order. If a position is written $i = jn + r$ with $0 \leq j < k$ and $1 \leq r \leq n$, its literal packet destination is

$$\pi_{k,n}(i) = kr - j. \quad (1)$$

The elementary identity

$$ki = kjn + kr \equiv kr - j \pmod{kn + 1}$$

shows that (1) is the nonzero representative of ki modulo M . Thus the packet operation and the residue map are the same map, while the two descriptions provide genuinely independent implementations.

2 All-period theorem

Theorem 1 (Fixed points and local periods). *For every $r \geq 1$,*

$$F_r := |\text{Fix}(\rho_{k,n}^r)| = \gcd(k^r - 1, M) - 1. \quad (2)$$

Moreover $\gcd(\rho(i), M) = \gcd(i, M)$ and, with $q_i = M / \gcd(i, M)$, the least period of i is

$$\tau(i) = \text{ord}_{q_i}(k). \quad (3)$$

Proof. Because $M \equiv 1 \pmod{k}$, k is a unit modulo M . A residue is fixed by the r th iterate exactly when $(k^r - 1)i \equiv 0 \pmod{M}$. This congruence has $\gcd(k^r - 1, M)$ solutions modulo M , one of which is the excluded zero, which proves (2). Multiplication by a unit preserves gcd with M . Dividing the congruence for a fixed position by $\gcd(i, M)$ leaves $k^r \equiv 1 \pmod{q_i}$; minimality gives (3). $\square$

Proposition 1 (Primitive cycles and source factors). *Let E_r be the number of positions of exact least period r and C_r the number of oriented cycles. Then*

$$E_r = \sum_{d|r} \mu(r/d) F_d, \quad C_r = E_r / r, \quad (4)$$

and $C_r = 0$ unless $r \mid \text{ord}_M(k)$. If P is the permutation matrix, the finite source factors are

$$Z_{k,n}(z) = \prod_{r \geq 1} (1 - z^r)^{-C_r}, \quad \det(\lambda I - P) = \prod_{r \geq 1} (\lambda^r - 1)^{C_r}. \quad (5)$$

Proof. Every fixed point has one least period, so $F_r = \sum_{d|r} E_d$ and Möbius inversion gives (4). Each cycle contributes exactly r positions. The cycle decomposition of a permutation matrix is a direct sum of cyclic shift blocks, whose zeta and characteristic factors are the two factors in (5). $\square$

References

- [1] J. Ellis, H. Fan, and J. Shallit, “The Cycles of the Multiway Perfect Shuffle Permutation,” *Discrete Mathematics & Theoretical Computer Science* 5 (2002). DOI: 10.46298/dmtcs.308.
- [2] R. W. Packard and E. S. Packard, “The Order of a Perfect k -Shuffle,” *The Fibonacci Quarterly* 32(2), 136–144 (1994). DOI: 10.1080/00150517.1994.12429237.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; the paper contains no target arithmetic, Euler-factor, target-zero, automorphy or Hilbert–Pólya claim. **Data and code.** Exact formulas, direct cycle ledgers and independent audit programs accompany HCS-C239. **AI-use disclosure.** Generative tools assisted drafting and code generation; all displayed claims are checked by the declared programs. This is not external peer review.